

Pilgrim Dot Farm Radio — Request Analysis

station.db requests table: 327 requests, 2026-09-30 → 2026-10-03

Automated analysis

2026-10-03 (UTC)

Snapshot & method

- **Source:** consistent read-only snapshot of ./pilgrim/station.db (SQLite VACUUM INTO, live DB untouched), saved to /tmp/station_copy.db (10.1 MB).
- **Scope:** the requests table (327 rows). Cross-referenced with items (2,181) and airplay (83,398).
- **Tooling:** Python 3.9 sqlite3 + matplotlib 3.9; report rendered with Pandoc 3.5 + Typst 0.12. All times UTC.

TL;DR

Metric	Value
Total requests	327 over 3.65 days (2026-09-30 00:38 → 2026-10-03 15:12)
Overall rate	3.7 / hour (median inter-arrival 3.0 min, mean 15.9 min)
Aired	241 (73.7 %) — p50 turnaround 21.1 min, p90 30.6 min, max 51.2 min
Rejected (moderation)	61 (18.7 %) — all closed near-instantly
Failed (3 attempts exhausted)	17 (5.2 %)
Evicted (FIFO cap)	7 (2.1 %) — all on 2026-10-02 02:18–02:29
Pending	1 (0.3 %) — “Down on the slop farm”, produced 11.2 min after request
Traffic	760 unique visitor IPs; 83,398 airplay events (61,102 of them on 2026-10-02)

1. When requests came in

1.1 Full-period time series

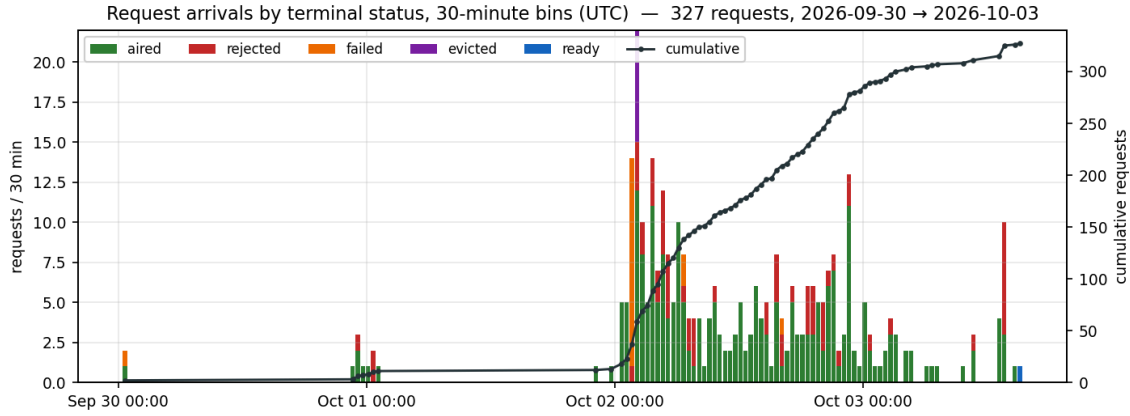


Figure 1: Request arrivals by terminal status, 30-minute bins

The window is dominated by a single burst: **2026-10-02 delivered 268 of 327 requests (82 %)**. The station went from a trickle (7 requests Sep 30, 6 on Oct 1) to a sustained flood on Oct 2, followed by a long tail of 46 requests on Oct 3 (snapshot cut off at 15:12).

Day	Total	Aired	Rejected	Failed	Evicted
2026-09-30	7	5	1	1	0
2026-10-01	6	4	2	0	0
2026-10-02	268	197	48	16	7
2026-10-03 (partial, →15:12)	46	35	10	0	0

1.2 Burst-day zoom

Burst-day zoom: 2026-10-02 (268 of 327 requests = 82% of all-time) — note the 01:00–03:00 UTC cluster incl. 14 production f

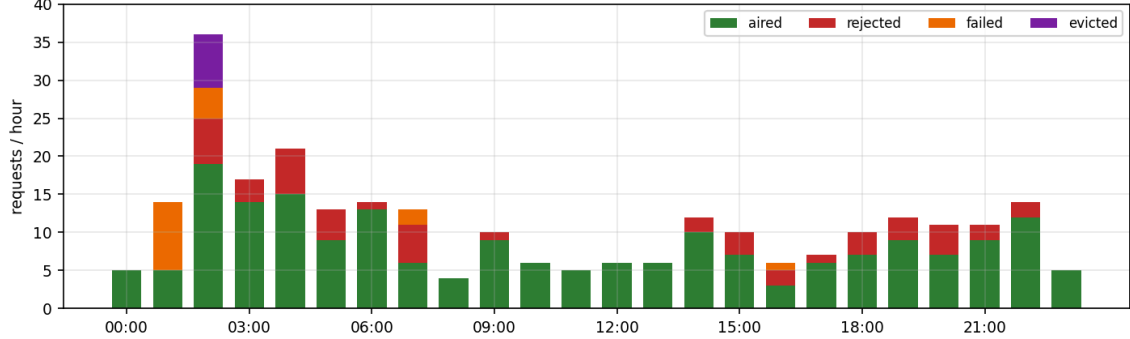


Figure 2: Burst day 2026-10-02

Oct 2 shows an intense early-UTC cluster (01:00–03:00: 67 requests, peaking at 36 in the 02:00 hour — the busiest hour of the whole window, 42 including other days). That is when the 17-attempt production failures and all 7 FIFO evictions happened (§3, §4). After ~03:00 the stream settles to a steady 4–14/hour for the rest of the day.

1.3 Hour-of-day cycle

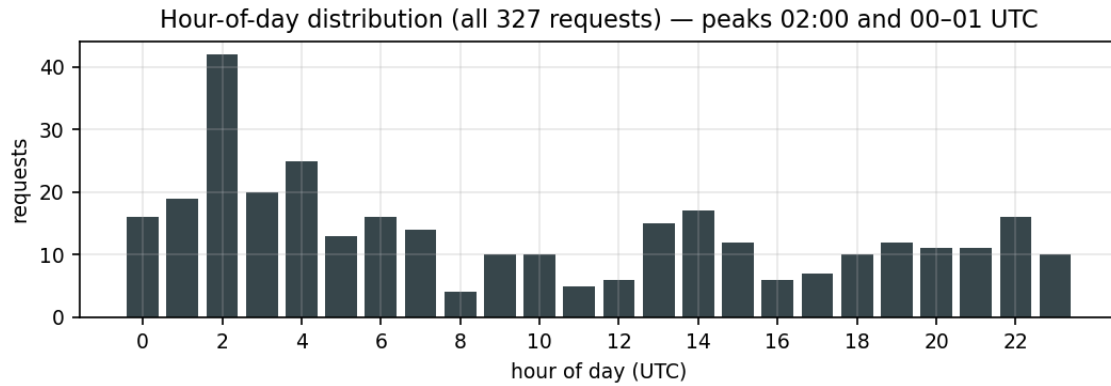


Figure 3: Hour-of-day distribution

Aggregated across all days, arrivals concentrate in the 00:00–05:00 UTC band (126 requests; ~2/3 of the total), with the single peak at 02:00 UTC. This reflects the one burst day more than a stable daily rhythm — pre-burst days (Sep 30, Oct 1) had only 13 requests total spread thinly.

2. Outcomes

2.1 Status breakdown

Status	n	%	Meaning
aired	241	73.7	song produced, QC-passed, aired
rejected	61	18.7	moderation rejected before production
failed	17	5.2	production failed after 3 attempts
evicted	7	2.1	dropped: request FIFO over cap
ready	1	0.3	produced, awaiting airplay (snapshot-time)

2.2 Rejection reasons (n = 61), grouped

Safety / policy category	n
Sexual / vulgar content	17
Defamation, insult, or stereotype (named persons, groups)	13
Unreadable / obfuscated / not a request	5
No contact info on the request line	5
Slop marker text (“song aboj...”)	3
Threatening	2
Unsafe activity	2
Drug use	2
Non-consent / “basement” scenarios	2
Domain / API-key-leak attempts	2
Social-media handles	2
Weapon	1
Prompt-injection attempt	1
Harassing toward the DJ	1
Profanity abbreviation	1
Crass bodily imagery	1
Removed by operator	1

Notable blocked items: a request **containing prompt-injection instructions to override moderation**, two that **mentioned a domain plus a leaked API key**, an “unreadable obfuscated symbol string”, and a real-person defamation (rejected as “*Defamatory allegation against a named person*” — id 5). Rejections close instantly (p50 = 0.0 min), so no production time is spent on them.

3. Latency and retries

request lifecycle latency (updated_at – created_at). Rejected/evicted close instantly (omitted).

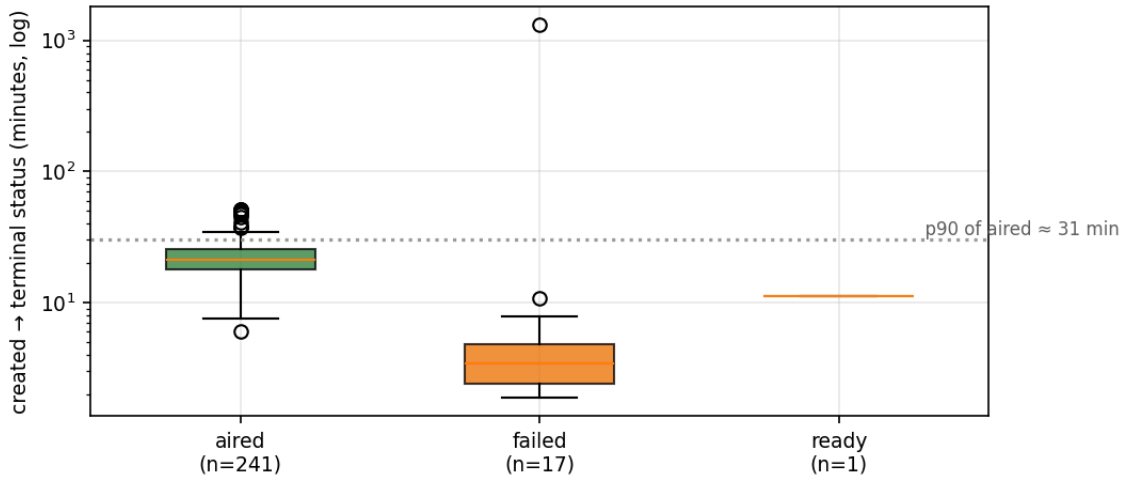


Figure 4: Lifecycle latency by status

- **Aired:** p50 21.1 min, p90 30.6 min, max 51.2 min from request to terminal state (brief → song generation → QC → normalize → aired). Consistent with the ~1.6 s compute-per-1 s-audio song-generation cost.
- **Failed:** p50 3.4 min (fast failures), but one outlier at 1,312 min: request #2 (“A rap song about sipping on syrup”, Sep 30 00:47) was only marked failed at 22:40 – a ~22-hour late sweep, likely the next-day slop/cleanup pass.
- **Retries:** of 241 aired requests, **226 (93.8 %) succeeded on the first attempt**; 15 needed 1–2 retries. All 17 failed requests exhausted exactly 3 attempts (attempts = retries spent; 0 = first try).

4. Production failure & eviction incident (2026-10-02)

Between **01:40 and 02:10 UTC** on Oct 2, **13 of the 17 failed requests** arrived, each failing within 2–5 minutes of creation (e.g. ids 25–38: “Hola”, “Th1stl3d0wn triumph...”, “Light me up (euro dance remix)”, “more cowbell”). The cadence — a rapid stream of short, evenly spaced failures — looks like a **~30-minute backend outage** (song generation) that the station absorbed without dropping the broadcast, per the “failing backend degrades inventory, never crashes the station” rule. Two more failures came later the same day (Tamil-language requests, 07:02–07:03; a goblin-groan request, 16:24), and one pre-burst failure on Sep 30 (#2).

Immediately after the failure cluster, during the peak of the burst, **7 requests were evicted** with reason `fifo over cap` (02:18–02:29 UTC) — the request queue hit its cap while the production pipeline was degraded. All 7 were dropped without being produced.

5. What the requests became

- **240 aired requests resolve to distinct song items** (424.9 min total, avg 106.2 s, range 31.6–148.1 s), plus DJ intros. Every aired request also got an `intro_item` (DJ talk-up).
- Most-played requested songs (from `items.play_count`):

Request (as submitted)	Aired title	Genre	Plays
“A song about football and CTE”	Concussion County	bluegrass	28
“Dirty south rap about biscuits and gravy”	Biscuits and Gravy (Two-Step Flow)	polka	26
“Ollie the dog loves to be wrong (bernedoodle remix)”	Ollie Loves to Be Wrong (Bernedoodle Remix)	chiptune	7
“A monastic chant about living off the sweat of one’s brow”	The Hymn of the Hon-est Brow	dungeon synth	7

Station context: 2,181 library items (470 songs, 580 DJ talk, 415 SFX, 397 intros, 159 field reports, 115 news; 9 retired), catalogue revision 394, 760 unique visitor IPs, 83,398 airplay events since the ledger opened 2026-10-01 22:04 UTC.

6. Data-quality observations

1. **Dangling FK:** request #1 (“A song about life being a turtle’s dream”, the very first request) points to `song_item_id = 68`, which no longer exists in `items` — the item was removed at some point without updating the request.
2. **reason doubles as a post-hoc note:** request #270 (“song abojut”) carries the slop-cleanup note “*slop: ‘song aboj’ marker (requests 268/269/270)*” yet has status `aired` — it produced item 1668 (“My Tractor’s Got a Polka Heart”), which aired 83×. Its siblings #268/#269 (submitted in the same 2-second window) were rejected. So the cleanup correctly flagged the slop text, but one already-produced duplicate still aired.
3. **attempts semantics:** 0 means “succeeded on first try” for aired requests, not “no attempt” — the field only records retry count.
4. **Airplay ledger starts 2026-10-01 22:04**, so airplay counts predate the oldest requests by nothing but postdate them by ~2 days; Oct 2’s 61,102 plays vs Oct 3’s 21,392 (partial day) matches the burst-day traffic.

Artifacts

- Snapshot: `/tmp/station_copy.db` (10.1 MB, includes WAL data; live DB unmodified)
- This report: `/tmp/analysis/report.md` → `/tmp/analysis/station_requests_report.pdf`
- Plots: `/tmp/analysis/fig1_arrivals.png`, `fig2_burst.png`, `fig3_latency.png`, `fig4_hour.png`